

Making a Whole

Australian Curriculum v9.0 | AC9M3N02

I am learning to: combine fractions with the same denominator to make one whole.

Name: _____ Date: _____

LEARN - keep the unit parts the same size

A whole strip has 5 equal parts. Start with $\frac{2}{5}$ and add the remaining $\frac{3}{5}$: $\frac{2}{5} + \frac{3}{5} = \frac{5}{5} = 1$ whole. The denominator stays 5.



2 shaded + 3 unshaded = 5 fifths

1. Shade the rest. Write the fraction you added.



a. $\frac{1}{4} + \underline{\hspace{1cm}} = \frac{4}{4}$



b. $\frac{3}{5} + \underline{\hspace{1cm}} = \frac{5}{5}$

2. Complete each whole. Use a fraction in every blank.

a. $\frac{1}{2} + \underline{\hspace{1cm}} = \frac{2}{2}$

b. $\frac{1}{3} + \underline{\hspace{1cm}} = \frac{3}{3}$

c. $\frac{7}{10} + \underline{\hspace{1cm}} = \frac{10}{10}$

d. $\frac{2}{4} + \underline{\hspace{1cm}} = \frac{4}{4}$

e. $\frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \underline{\hspace{1cm}} = \frac{5}{5}$

3. A strip has 10 equal lengths. Shade $\frac{4}{10}$.

Fraction left: _____

$\frac{4}{10} + \underline{\hspace{1cm}} = 1$

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I am learning to: find the missing part and explain how parts form a whole.

4. APPLY - the whole collection has 12 counters.

Mia has $\frac{1}{3}$ of the counters. Leo has all the rest. What fraction does Leo have? How many counters does each child have?

5. THINK - should the denominators be added?

Ben says, " $\frac{2}{5} + \frac{3}{5} = \frac{5}{10}$." Use the strip to explain his mistake.



6. CHALLENGE - find two different pairs of tenths.

Each pair must make one whole. Both fractions must be more than zero. Order swaps do not count as a new pair.

$$\underline{\quad} + \underline{\quad} = 10/10$$

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7. Explain why $\frac{3}{3}$ represents one whole.

REVIEW - tick what you can do

☐ I find the missing part.

☐ I explain one whole.

☐ I keep the denominator.

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I am learning to: check my answers and explain my thinking.

COMPLETE ANSWER KEY

1. a. Shade the remaining 3 parts: add $\frac{3}{4}$. b. Shade the remaining 2 parts: add $\frac{2}{5}$.
2. a. $\frac{1}{2}$. b. $\frac{2}{3}$. c. $\frac{3}{10}$. d. $\frac{2}{4}$. e. $\frac{1}{5}$. In each case all equal parts of the whole are now included.
3. Shade any 4 of the 10 equal lengths. The fraction left is $\frac{6}{10}$; $\frac{4}{10} + \frac{6}{10} = 1$.
4. Mia has 4 counters ($\frac{1}{3}$). Leo has 8 counters ($\frac{2}{3}$). Three equal groups of 4 make the collection of 12.
5. The pieces are still fifths. There are five fifths altogether: $\frac{2}{5} + \frac{3}{5} = \frac{5}{5} = 1$. Adding the parts does not split the whole into 10 pieces.
6. Example: $\frac{1}{10} + \frac{9}{10} = \frac{10}{10}$ and $\frac{4}{10} + \frac{6}{10} = \frac{10}{10}$. Other pairs: $\frac{2}{10} + \frac{8}{10}$, $\frac{3}{10} + \frac{7}{10}$, $\frac{5}{10} + \frac{5}{10}$. Accept any two different valid pairs; do not count reversed order twice.
7. The whole has three equal parts. Three thirds include every part, so $\frac{3}{3}$ is one whole.

Review. Student self-assessment; no fixed answers.

VISUAL ANSWER EXAMPLE



$$\frac{2}{5} + \frac{3}{5} = \frac{5}{5} = 1$$

Two fifths and three fifths complete the whole.

TEACHER / PARENT NOTE

This is part-whole completion, not a general fraction addition lesson. Use strips or counters before symbols. If a child adds denominators, point to the unchanged unit part and ask its name. Check that the child can explain why all the equal parts together form one whole.

Checked: 3 pages • all answers • equal models • AC9M3N02